Manage S3 audit configurations

ONTAP 9.10.1 REST API Documentation

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Manage S3 audit configurations

Protocols audit svm.uuid object-store endpoint overview

Overview

S3 events auditing is a security measure that enables you to track and log certain S3 events on storage virtual machines (SVMs). You can track potential security problems and provides evidence of any security breaches.

Examples

Creating an S3 audit entry with log rotation size and log retention count

To create an S3 audit entry with log rotation size and log retention count, use the following API. Note the *return_records=true* query parameter is used to obtain the newly created entry in the response.

```
# The API:
POST /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{ \"enabled\":
true, \"events\": { \"data\": false, \"management\": false}, \"log\": {
\"format\": \"json\", \"retention\": { \"count\": 10 }, \"rotation\": {
\"size\": 2048000 }}, \"log_path\": \"/\"}"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "enabled": true,
    "events": {
      "data": false,
      "management": false
    },
    "log": {
      "format": "json",
      "rotation": {
        "size": 2048000
      },
      "retention": {
        "count": 10,
        "duration": "0s"
      }
    },
    "log_path": "/"
  }
],
"num_records": 1
}
```

Creating an S3 audit entry with log rotation schedule and log retention duration

To create an S3 audit entry with log rotation schedule and log retention duration, use the following API. Note that the *return_records=true* query parameter is used to obtain the newly created entry in the response.

```
# The API:
POST /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/audit/a8d64674-13fc-11e9-
87b1-005056a7ae7e/object-store?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{ \"enabled\":
false, \"events\": { \"data\": true, \"management\": true }, \"log\": {
\"format\": \"json\", \"retention\": { \"duration\": \"P4DT12H30M5S\" },
\"rotation\": { \"schedule\": { \"days\": [1, 5, 10, 15], \"hours\": [0,
1, 6, 12, 18, 23], \"minutes\": [10, 15, 30, 45, 59], \"months\": [0],
\"weekdays\": [0, 2, 5] } } }, \"log_path\": \"/\"}"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "enabled": true,
    "events": {
      "data": true,
      "management": true
    },
    "log": {
      "format": "json",
      "rotation": {
        "schedule": {
          "minutes": [
            10,
            15,
            30,
            45,
            59
          ],
          "hours": [
            0,
            1,
            6,
            12,
            18,
            23
          ],
```

```
          "weekdays": [
            0,
            2,
            5
          ],
          "days": [
            1,
            5,
            10,
            15
          ],
          "months": [
            0
          ]
        }
      },
      "retention": {
        "count": 0,
        "duration": "P4DT12H30M5S"
      }
    },
    "log_path": "/"
  }
],
"num_records": 1
}
```

Retrieving an S3 audit configuration for all SVMs in the cluster

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/*/object-
store?fields=*&return_records=true&return_timeout=15" -H "accept:
application/json"

# The response:
{
"records": [
  {
    "svm": {
```

```
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "enabled": true,
    "events": {
      "data": false,
      "management": false
    },
    "log": {
      "format": "json",
      "rotation": {
        "size": 2048000
      },
      "retention": {
        "count": 10,
        "duration": "0s"
      }
    },
    "log_path": "/"
  },
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "enabled": true,
    "events": {
      "data": true,
      "management": true
    },
    "log": {
      "format": "json",
      "rotation": {
        "schedule": {
          "minutes": [
            10,
            15,
            30,
            45,
            59
          ],
          "hours": [
            0,
            1,
            6,
            12,
```

```
            18,
            23
          ],
          "weekdays": [
            0,
            2,
            5
          ],
          "days": [
            1,
            5,
            10,
            15
          ],
          "months": [
            0
          ]
        }
      },
      "retention": {
        "count": 0,
        "duration": "P4DT12H30M5S"
      }
    },
    "log_path": "/"
  }
],
"num_records": 2
}
```

Retrieving specific entries with event list as data and management event for an SVM

The configuration returned is identified by the events in the list of S3 audit configurations of an SVM.

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/*/object-
store?events.data=true&events.management=true&return_records=true&return_t
imeout=15" -H "accept: application/json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "events": {
      "data": true,
      "management":true
    }
  },
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "events": {
      "data": true,
      "management": true
    }
  }
],
"num_records": 2
}
```

Retrieving a specific S3 audit configuration of an SVM

The configuration returned is identified by the UUID of its SVM.

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store/" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
  "name": "vs1"
},
"enabled": true,
"events": {
  "data": false,
  "management": false
},
"log": {
  "format": "json",
  "rotation": {
    "size": 2048000
  },
  "retention": {
    "count": 10,
    "duration": "0s"
  }
},
"log_path": "/"
}
```

Updating a specific S3 audit configuration of an SVM

The configuration is identified by the UUID of its SVM and the provided information is updated.

```
# The API:
PATCH /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X PATCH "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store/" -H "accept: application/json" -H
"Content-Type: application/json" -d "{ \"enabled\": false}"
```

Deleting a specific S3 audit configuration of an SVM

The entry to be deleted is identified by the UUID of its SVM.

```
# The API:
DELETE /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X DELETE "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store" -H "accept: application/json"
```

Delete an S3 audit configuration

DELETE `/protocols/audit/{svm.uuid}/object-store`

Introduced In: 9.10

Deletes an S3 audit configuration.

Related ONTAP commands

- `vserver object-store-server audit disable`
- `vserver object-store-server audit delete`

Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

Parameters

Name	Type	In	Required	Description
force	boolean	query	False	Indicates whether a force deletion of the audit configuration is enabled.
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
140902420	Failed to delete audit configuration for the SVM.
140902421	Failed to delete audit configuration for the SVM because audit is enabled for the SVM.
140902422	Failed to delete audit configuration for the SVM because final conolidation is in progress. Wait a few mintues, and try the operation again.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve S3 audit configurations

GET `/protocols/audit/{svm.uuid}/object-store`

Introduced In: 9.10

Retrieves S3 audit configurations.

Related ONTAP commands

- `vserver object-store-server audit show`

Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

Parameters

Name	Type	In	Required	Description
enabled	boolean	query	False	Filter by enabled
log_path	string	query	False	Filter by log_path

Name	Type	In	Required	Description
log.retention.count	integer	query	False	Filter by log.retention.count
log.retention.duration	string	query	False	Filter by log.retention.duration
log.rotation.schedule.months	integer	query	False	Filter by log.rotation.schedule.months
log.rotation.schedule.days	integer	query	False	Filter by log.rotation.schedule.days
log.rotation.schedule.hours	integer	query	False	Filter by log.rotation.schedule.hours
log.rotation.schedule.minutes	integer	query	False	Filter by log.rotation.schedule.minutes
log.rotation.schedule.weekdays	integer	query	False	Filter by log.rotation.schedule.weekdays
log.rotation.size	integer	query	False	Filter by log.rotation.size
log.format	string	query	False	Filter by log.format
svm.name	string	query	False	Filter by svm.name
events.management	boolean	query	False	Filter by events.management
events.data	boolean	query	False	Filter by events.data
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.
fields	array[string]	query	False	Specify the fields to return.

Name	Type	In	Required	Description
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
enabled	boolean	Specifies whether or not auditing is enabled on the SVM.
events	events	
log	s3_log	

Name	Type	Description
log_path	string	The audit log destination path where consolidated audit logs are stored.
svm	svm	

Example response


```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

events

Name	Type	Description
data	boolean	Data events
management	boolean	Management events

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

retention

Name	Type	Description
count	integer	Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration".
duration	string	Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count".

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

Name	Type	Description
days	array[integer]	Specifies the day of the month schedule to rotate audit log. Leave empty for all.

Name	Type	Description
hours	array[integer]	Specifies the hourly schedule to rotate audit log. Leave empty for all.
minutes	array[integer]	Specifies the minutes schedule to rotate the audit log.
months	array[integer]	Specifies the months schedule to rotate audit log. Leave empty for all.
weekdays	array[integer]	Specifies the weekdays schedule to rotate audit log. Leave empty for all.

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

Name	Type	Description
now	boolean	Manually rotates the audit logs. Optional in PATCH only. Not available in POST.
schedule	audit_schedule	Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.
size	integer	Rotates logs based on log size in bytes.

s3_log

Name	Type	Description
_links	_links	

Name	Type	Description
format	string	Format in which the logs are generated by the consolidation process. Possible values are: • json - ONTAP-specific Json log format. • Default value: 1 • enum: ["json"] • Introduced in: 9.10
retention	retention	
rotation	rotation	Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments

Name	Type	Description
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update the S3 audit configuration for an SVM

PATCH `/protocols/audit/{svm.uuid}/object-store`

Introduced In: 9.10

Updates an S3 audit configuration for an SVM.

Important notes

- `events` - Not specifying either data or management is equivalent to setting it to false.

Related ONTAP commands

- `vserver object-store-server audit modify`

Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

Parameters

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Request Body

Name	Type	Description
enabled	boolean	Specifies whether or not auditing is enabled on the SVM.
events	events	
log	s3_log	

Name	Type	Description
log_path	string	The audit log destination path where consolidated audit logs are stored.
svm	svm	

Example request

```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
140902401	Failed to create an audit configuration for the SVM.
140902402	Audit configuration is already present.
140902402	Audit configuration is already enabled.
140902403	Failed to create staging volume.
140902415	Failed to modify an audit configuration because no audit configuration exists for SVM.
140902416	Failed to modify audit configuration for SVM.
140902422	Final consolidation is in progress, audit delete failed.
140902423	Failed to delete the audit configuration for the SVM.
140902425	Audit configuration is not available for disabling.
140902430	Audit configuration is not available for enabling.
140902431	Audit enable failed, audit configuration already enabled for the SVM.
140902432	Final consolidation is in progress, audit enable failed.
140902445	Audit disable failed, audit configuration does not exist for the SVM.
140902446	Audit configuration is already disabled.
140902446	Audit disable failed, audit configuration does not exist for the SVM.
140902456	The specified log_path does not exist.
140902457	The log_path must be a directory.
140902458	The log_path must be a canonical path in the SVM's namespace.
140902459	The log_path cannot be empty.
140902460	Rotate size must be greater than or equal to 1024 KB.
140902461	The destination path must not contain a symbolic link.
140902470	The log_path exceeds a maximum supported length of characters.
140902471	The log_path contains an unsupported read-only (DP/LS) volume.
140902472	The log_path is not a valid destination for the SVM.

Error Code	Description
140902474	The log_path contains an unsupported Snaplock volume.
140902478	The log_path validation failed.
140902478	The log_path cannot be accessed for validation.
140902490	Audit configuration is absent for rotate.
140902491	Failed to rotate audit log.
140902492	Cannot rotate audit log, auditing is not enabled for this SVM.

ONTAP Error Response Codes

Error Code	Description
9699340	SVM UUID lookup failed
9699407	Additional fields are provided

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

events

Name	Type	Description
data	boolean	Data events
management	boolean	Management events

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

retention

Name	Type	Description
count	integer	Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration".
duration	string	Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count".

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

Name	Type	Description
days	array[integer]	Specifies the day of the month schedule to rotate audit log. Leave empty for all.

Name	Type	Description
hours	array[integer]	Specifies the hourly schedule to rotate audit log. Leave empty for all.
minutes	array[integer]	Specifies the minutes schedule to rotate the audit log.
months	array[integer]	Specifies the months schedule to rotate audit log. Leave empty for all.
weekdays	array[integer]	Specifies the weekdays schedule to rotate audit log. Leave empty for all.

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

Name	Type	Description
now	boolean	Manually rotates the audit logs. Optional in PATCH only. Not available in POST.
schedule	audit_schedule	Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.
size	integer	Rotates logs based on log size in bytes.

s3_log

Name	Type	Description
_links	_links	

Name	Type	Description
format	string	Format in which the logs are generated by the consolidation process. Possible values are: • json - ONTAP-specific Json log format. • Default value: 1 • enum: ["json"] • Introduced in: 9.10
retention	retention	
rotation	rotation	Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

s3_audit

Auditing for NAS events is a security measure that enables you to track and log certain S3 events on SVMs.

Name	Type	Description
enabled	boolean	Specifies whether or not auditing is enabled on the SVM.
events	events	
log	s3_log	

Name	Type	Description
log_path	string	The audit log destination path where consolidated audit logs are stored.
svm	svm	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create an S3 audit configuration

POST `/protocols/audit/{svm.uuid}/object-store`

Introduced In: 9.10

Creates an S3 audit configuration.

Required properties

- `log_path` - Path in the owning SVM namespace that is used to store audit logs.

Default property values

If not specified in POST, the following default property values are assigned:

- `enabled` - *true*
- `events.data` - *true*

- `events.management` - *false*
- `log.format` - *json*
- `log.retention.count` - *0*
- `log.retention.duration` - *PT0S*
- `log.rotation.size` - *100MB*
- `log.rotation.now` - *false*

Related ONTAP commands

- `vserver object-store-server audit create`
- `vserver object-store-server audit enable`

Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

Parameters

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Request Body

Name	Type	Description
enabled	boolean	Specifies whether or not auditing is enabled on the SVM.
events	events	
log	s3_log	

Name	Type	Description
log_path	string	The audit log destination path where consolidated audit logs are stored.
svm	svm	

Example request

```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

Response

```
Status: 202, Accepted
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[s3_audit]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "log": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "format": "json",
      "retention": {
        "duration": "P4DT12H30M5S"
      },
      "rotation": {
        "schedule": {
          "days": {
          },
          "hours": {
          },
          "minutes": {
          },
          "months": {
          },
          "weekdays": {
          }
        }
      }
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
```

```
    }
  }
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
140902401	Failed to create an audit configuration for the SVM.
140902402	Audit configuration is already present.
140902402	Audit configuration is already enabled.
140902403	Failed to create staging volume.
140902415	Failed to modify an audit configuration because no audit configuration exists for the SVM.
140902416	Failed to modify audit configuration for SVM.
140902422	Final consolidation is in progress, audit delete failed.
140902423	Failed to delete the audit configuration for the SVM.
140902425	Audit configuration is not available for disabling.
140902430	Audit configuration is not available for enabling.
140902431	Audit enable failed, audit configuration already enabled for the SVM.
140902432	Final consolidation is in progress, audit enable failed.
140902445	Audit disable failed, audit configuration does not exist for the SVM.
140902446	Audit disable failed, audit configuration does not exist for the SVM.
140902447	Audit disable failed.
140902456	The specified log_path does not exist.
140902457	The log_path must be a directory.
140902458	The log_path must be a canonical path in the SVM's namespace.
140902459	The log_path cannot be empty.
140902460	Rotate size must be greater than or equal to 1024 KB.
140902461	The destination path must not contain a symbolic link.

Error Code	Description
140902470	The log_path exceeds a maximum supported length of characters.
140902471	The log_path contains an unsupported read-only (DP/LS) volume.
140902472	The log_path is not a valid destination for the SVM.
140902474	The log_path contains an unsupported Snaplock volume.
140902478	The log_path validation failed.
140902478	The log_path cannot be accessed for validation.
140902490	Audit configuration is absent for rotate.
140902491	Failed to rotate audit log.
140902492	Cannot rotate audit log, auditing is not enabled for this SVM.

ONTAP Error Response Codes

Error Code	Description
9699340	SVM UUID lookup failed
9699407	Additional fields are provided

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

events

Name	Type	Description
data	boolean	Data events
management	boolean	Management events

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

retention

Name	Type	Description
count	integer	Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration".
duration	string	Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count".

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

Name	Type	Description
days	array[integer]	Specifies the day of the month schedule to rotate audit log. Leave empty for all.

Name	Type	Description
hours	array[integer]	Specifies the hourly schedule to rotate audit log. Leave empty for all.
minutes	array[integer]	Specifies the minutes schedule to rotate the audit log.
months	array[integer]	Specifies the months schedule to rotate audit log. Leave empty for all.
weekdays	array[integer]	Specifies the weekdays schedule to rotate audit log. Leave empty for all.

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

Name	Type	Description
now	boolean	Manually rotates the audit logs. Optional in PATCH only. Not available in POST.
schedule	audit_schedule	Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.
size	integer	Rotates logs based on log size in bytes.

s3_log

Name	Type	Description
_links	_links	

Name	Type	Description
format	string	Format in which the logs are generated by the consolidation process. Possible values are: • json - ONTAP-specific Json log format. • Default value: 1 • enum: ["json"] • Introduced in: 9.10
retention	retention	
rotation	rotation	Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

s3_audit

Auditing for NAS events is a security measure that enables you to track and log certain S3 events on SVMs.

Name	Type	Description
enabled	boolean	Specifies whether or not auditing is enabled on the SVM.
events	events	
log	s3_log	

Name	Type	Description
log_path	string	The audit log destination path where consolidated audit logs are stored.
svm	svm	

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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